DINESH KUMAR S

AI Engineer • GenAI & Agentic Systems • NLP & RAG Specialist

dineshsuresh2903@gmail.com | 07708765012 | linkedin.com/in/dinesh-kumar-s-00036b280 | dineshkumar8727.github.io/Portfolio/

PROFESSIONAL SUMMARY

Results-driven AI Engineer with 1+ year of hands-on experience designing and deploying production-grade Generative AI, LLM, and Agentic AI systems for large-scale financial document intelligence. Proven expertise in RAG architectures, NLP extraction pipelines, and multi-modal document processing. Adept at bridging ML research and real-world engineering — from semantic embeddings and vector search to automated multi-step agentic workflows. Passionate about building reliable, grounded AI systems that reduce hallucinations and drive measurable business value.

EXPERIENCE

Software Developer — Data Mining & AI | Factentry Data Solutions Pvt. Ltd. (SIX Group) *June 2025 – Present*

- Architected and deployed ML-driven NLP extraction pipelines and RAG systems using semantic embeddings and vector search, enabling accurate retrieval-augmented generation over large-scale financial PDFs.
- Implemented query-time retrieval mechanisms to ground LLM responses, significantly reducing hallucinations in production.
- Built Agentic AI workflows on the n8n automation platform to orchestrate multi-step decision-making, automated email reporting, and intelligent content generation.
- Designed end-to-end GenAI-powered document intelligence solutions integrating OCR, LLMs, and semantic search for structured financial data extraction at scale.

Graduate Intern — Data Science & MLOps | Factentry Data Solutions Pvt. Ltd. (SIX Group) *Nov 2024 – May 2025*

- Engineered multilingual PDF analysis pipelines integrating OCR, regex, language detection, and LLMs — supporting structured extraction from noisy, real-world documents.
- Developed advanced image-based document clustering using ResNet, VGG16, and Vision Transformers (ViT), combined with PCA + DBSCAN and 3D visualization for document triage.
- Built a scalable multi-modal PDF Q&A chatbot (FastAPI + Streamlit + Groq LLaMA-70B + LangChain) with NER, conversational memory, and page-level citation support.

KEY PROJECTS

GenAI-Powered RAG-Based Document Extraction System

- Designed a production-grade RAG pipeline transforming unstructured financial PDFs into structured, analytics-ready data using semantic chunking, embedding generation, and vector search.
- Architected context-aware retrieval with grounded LLM responses; integrated Agentic AI for multi-step reasoning, validation, and field-level extraction at scale.
- Tech: LangChain, FAISS / vector DB, OpenAI / Groq LLMs, FastAPI, Python

ML-Driven NLP Extraction System

- Built TF-IDF and probabilistic ML classifiers for high-precision unstructured document extraction with sliding-window context analysis and adjacent-sentence merging for noisy text.
- Deployed serialized vectorizers and classifiers (pickle) into a hybrid ML + rule-based production inference pipeline with confidence-based decision thresholds.
- Tech: Scikit-learn, NLTK, spaCy, Regex, Python

GenAI Conversational Agent for Fixed Income Analysis

- Developed a conversational PDF Q&A agent integrating Mistral and LLaMA-70B (via Groq API) with Streamlit for fixed income document analysis.

- Implemented table extraction, metadata analysis, multilingual summarization, page-level referencing, NER, and conversational memory using LangChain and PDFPlumber.
- Tech: LangChain, Groq, Streamlit, PDFPlumber, NLTK, Prompt Engineering

GenAI & Agentic AI Email Reporting Automation (n8n)

- Designed an end-to-end autonomous email reporting system on the n8n platform with multi-step AI agents that analyze data, generate contextual reports, and dispatch personalized emails.
- Integrated LLMs, external APIs, and data sources within n8n to enable intelligent summarization, validation, and scheduled delivery — fully automated.
- Tech: n8n, LLM APIs, GenAI, Workflow Automation

TECHNICAL SKILLS

AI / GenAI	LLMs, RAG, Agentic AI, Prompt Engineering, Semantic Search, Vector DBs
ML / NLP	NLP, TF-IDF, Text Classification, NER, Embeddings, Image Processing
Frameworks	LangChain, FastAPI, Streamlit, n8n, Hugging Face, Scikit-learn
Models	LLaMA-70B, Mistral, ResNet, VGG16, Vision Transformers (ViT)
Languages	Python, SQL
Tools & Cloud	Power BI, Microsoft Azure, EDA, Git, Docker (basic)

EDUCATION

M.Tech Integrated — CSE, Specialization: Data Science *2025*

Vellore Institute of Technology (VIT), Vellore • CGPA: 7.68 / 10

CERTIFICATIONS

Microsoft Azure AI Fundamentals — Microsoft (2023)
Applied Data Science — Smart Internz (2023)